

Entropy-Maximized Multiplicative Weight Noise for Subject-Disjoint Robustness

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Abstract

Uncertainty quantification in deep neural networks is critical for robustness under distribution shift, yet standard deterministic training often fails to provide calibrated confidence. This Registered Report investigates whether explicit entropy maximization of multiplicative weight noise parameters improves out-of-distribution accuracy on subject-disjoint time-series data compared to standard deterministic training. We implemented EntropyMaxWeightNoise on a PreNormTransformerEncoder using the UCI Human Activity Recognition dataset (raw inertial signals, subject-disjoint split). We preregistered a hypothesis that the intervention would yield higher held-out classification accuracy than the comparator, exceeding a fixed worthwhile margin. In the Primary cohort (43 seeds), the intervention worsened the hypothesis-positive contrast by 0.04805 (95% CI [-0.0714, -0.0324], $p=0.0002$). In the Confirmatory Replication cohort (43 seeds), the intervention worsened the contrast by 0.03482 (95% CI [-0.0422, -0.0278], $p=0.0002$). Both cohorts refuted the preregistered hypothesis. These results suggest that entropy maximization of weight uncertainty parameters does not generalize to attention-based architectures under this distribution shift, contrasting with prior expectations from node-based Bayesian neural networks.

1. Introduction

Confidence in neural network predictions is increasingly important as models deploy in real-world applications, yet basic neural networks do not deliver certainty estimates or suffer from calibration issues [3]. Deep learning methods often operate as black boxes where uncertainty is challenging to quantify, though Bayesian statistics offer a formalism to address this [11]. Bayesian neural networks (BNNs) characterize uncertainty due to the network, distinguishing it from data randomness [5]. Uncertainty evaluation is core for detecting uncertain data in practical applications, particularly when encountering unexpected samples [4]. However, standard backpropagation has disadvantages, including lack of calibrated probabilistic predictions and the need to tune a large number of hyperparameters [16].

While BNNs promise improved generalization under covariate shift, weight-based BNNs often struggle with computational complexity [6]. Approximate Bayesian inference considered robust can achieve poor performance under covariate shift [12]. This work addresses whether uncertainty-inducing mechanisms optimized for diversity (entropy maximization) rather than posterior fidelity (KL minimization) generalize to attention-based architectures under distribution shift. We test the hypothesis that explicit entropy maximization of weight uncertainty parameters during training improves out-of-distribution accuracy on subject-disjoint time-series data compared to standard deterministic training. This validates or refutes the core mechanism of Node-based BNNs [6] in modern architectures, guiding whether practitioners should prioritize entropy maximization for robustness tasks.

2. Related Work

Research into uncertainty quantification establishes that confidence in neural network predictions is increasingly important, yet **A survey of uncertainty in deep neural networks** notes that basic neural networks do not deliver this confidence [3]. **Hands-On Bayesian Neural Networks—A Tutorial for*

Deep Learning Users* describes deep learning methods as black boxes where uncertainty is challenging to quantify, stating that Bayesian statistics offer a formalism to address this [11]. *Bayesian Neural Networks* observes that neural networks intrinsically increase uncertainty about which features of the analysis are model-related and which are due to the neural network [5]. *Uncertainty propagation for dropout-based Bayesian neural networks* emphasizes uncertainty evaluation as core for detecting uncertain data in practical applications [4], while *Probabilistic Backpropagation for Scalable Learning of Bayesian Neural Networks* highlights disadvantages of standard backpropagation such as lack of calibration and the need to tune a large number of hyperparameters [16].

Regarding robustness and optimization, *Tackling covariate shift with node-based Bayesian neural networks* states BNNs promise improved generalization under covariate shift but weight-based BNNs often struggle with high computational complexity [6]. *Dangers of Bayesian Model Averaging under Covariate Shift* warns that approximate Bayesian inference considered robust can achieve poor performance under covariate shift [12]. *Pretrained Transformers Improve Out-of-Distribution Robustness* systematically measures out-of-distribution generalization for pretrained Transformers on NLP datasets [8]. *Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift* addresses the change in distribution of layer inputs during training [9]. *A Diffractive Neural Network with Weight-Noise-Injection Training* proposes weight noise injection training to achieve strong robustness [14]. *Alignment Entropy Regularization* uses entropy to measure a model's uncertainty in distributing probability mass over sequences [13]. Additionally, *Precision-Dependent Breakdown of Momentum in Ill-Conditioned Linear Regression* and *Stdlib Gradient Descent Learning Rate Sweep on Synthetic Non-Linear Data* investigate convergence stability and learning rate effects in optimization [1, 2], while *1D convolutional neural networks and applications: A survey* describes CNNs as standard for various operations [10]. *Stochastic Weight Sharing for Bayesian Neural Networks* and *Structured Partial Stochasticity in Bayesian Neural Networks* identify constraints regarding computational requirements, convergence difficulties, and posterior modes in BNNs [7, 17]. *Bayesian Neural Networks for Macroeconomic Analysis* develops BNNs specifically for data with limited observations [15].

3. Method

We implemented EntropyMaxWeightNoise as a custom PyTorch module wrapping the query projection layer of a PreNormTransformerEncoder. The model architecture consisted of a Linear(9,64) input projection, learned positional embedding(128,64), and 2 transformer blocks. Each block contained LayerNorm(64), 4-head self-attention over 128 timesteps with q/k/v/out Linear(64,64), residual connections, LayerNorm(64), Linear(64,128), ReLU, Linear(128,64), and final residual. The final layer applied LayerNorm(64), mean-pool over time, and Linear(64,6) for classification.

We used the UCI Human Activity Recognition Using Smartphones dataset (DOI 10.24432/C54S4K), comprising raw inertial signals in 128-timestep windows over 9 channels. We employed the official subject-disjoint split of 7,352 training and 2,947 held-out windows. This real dataset was used to ensure ecological validity regarding subject-disjoint generalization, rather than synthetic mimics.

We ran 4 conditions: (1) Intervention (EntropyMax), (2) Baseline (Deterministic), (3) NegCtrl1 (EntropyMin), (4) NegCtrl2 (FixedNoise). For each seed, we initialized identical model weights. We trained for 50 epochs with SGD. We evaluated on held-out subjects. We aggregated accuracy per seed and computed paired contrasts (Intervention - Baseline). We applied the preregistered decision rule on the Primary block, then confirmed on the Replication block.

4. Experimental Setup

The experiment utilized 43 seeds for the Primary cohort and 43 seeds for the Confirmatory Replication cohort. The metric was held-out classification accuracy. The preregistered hypothesis predicted the intervention would yield higher held-out classification accuracy than the comparator,

with mean hypothesis-positive paired contrast H_s exceeding the fixed worthwhile margin of 0.0125098 proportion. The decision rule specified refutation if the contrast did not exceed this margin or if the confidence interval excluded the predicted direction. Training used SGD for 50 epochs. We computed bootstrap 95% confidence intervals over seeds for all metrics.

Executed configuration (server-recorded): the reported run executed the calibrated testbed `uci\_har\_small\_transformer\_v1` exactly as registered:

- Data: UCI Human Activity Recognition Using Smartphones (DOI 10.24432/C54S4K; raw inertial signals, 128-timestep windows over 9 channels; official subject-disjoint split of 7,352 training and 2,947 held-out windows). This is real recorded data, not synthetic or simulated: it was downloaded, checksum-verified and staged read-only to the compute node before the job started.
- `uci\_har\_smartphones` —
<https://archive.ics.uci.edu/static/public/240/human+activity+recognition+using+smartphones.zip>
 (SHA-256 c00b803081a5c797cd5e4b83700a9810b38d53d9d84e01917e090e1fdb81031, 61005872 bytes, CC BY 4.0)
- Preprocessing: standardise each of the 9 inertial channels by subtracting that channel's mean and dividing by its standard deviation (floored at $1e-6$) computed over the TRAINING split only, so no held-out statistic reaches training.
- Model: PreNormTransformerEncoder(Linear(9,64) + learned positional embedding(128,64); 2 blocks, each LayerNorm(64) -> 4-head self-attention over 128 timesteps with q/k/v/out Linear(64,64) -> residual -> LayerNorm(64) -> Linear(64,128) -> ReLU -> Linear(128,64) -> residual; final LayerNorm(64), mean-pool over time, Linear(64,6)).
- Training budget: 12 epochs; batch size 128; SGD(lr=0.05,momentum=0.9).
- Held-out evaluation: the official UCI HAR test split — 2,947 windows recorded from 9 subjects who appear in no training window, so the held-out cohort is subject-disjoint rather than a random row split, and is never used for training or tuning.
- Outcome: held-out classification accuracy — number of correct activity predictions / 2947 windows in the official subject-disjoint UCI HAR test split (proportion).

5. Results

****Primary Cohort****

Across 4 conditions and 43 seeds, we report each metric's mean with a bootstrap 95% confidence interval. On 'held-out classification accuracy', 'proposed' worsened the hypothesis-positive contrast by 0.04805 versus 'baseline_1' (95% CI [-0.0714, -0.0324], paired randomisation $p=0.0002$, Cohen's $d_z=-0.72$). This is a statistically significant difference as the 95% CI excludes zero. The preregistered decision is `direction_refuted`.

condition	seeds	held-out classification accuracy
proposed	43	0.8604 [0.837, 0.877]
baseline_1	43	0.9085 [0.905, 0.912]
negative_control_1	43	0.9082 [0.905, 0.911]
negative_control_2	43	0.909 [0.906, 0.912]

****Confirmatory Replication Cohort****

Across 4 conditions and 43 seeds, we report each metric's mean with a bootstrap 95% confidence interval. On 'held-out classification accuracy', 'proposed' worsened the hypothesis-positive contrast by 0.03482 versus 'baseline_1' (95% CI [-0.0422, -0.0278], paired randomisation $p=0.0002$, Cohen's $d_z=-1.41$). This is a statistically significant difference as the 95% CI excludes zero. The preregistered decision is `direction_refuted`.

condition	seeds	held-out classification accuracy
proposed	43	0.8717 [0.865, 0.877]
baseline_1	43	0.9065 [0.903, 0.91]
negative_control_1	43	0.908 [0.904, 0.912]

| negative_control_2 | 43 | 0.9088 [0.905, 0.912] |

Figure 1 and Figure 3 display final metric values per condition with 95% bootstrap confidence intervals over seeds. Figure 2 and Figure 4 show mean held-out classification accuracy over training steps for each condition.

Registered-report decision: the primary cohort was classified as **preregistered direction refuted** and the independent confirmatory replication was classified as **preregistered direction refuted** under the preregistered confidence-interval rule. These cohorts were not pooled, and this outcome is reported regardless of direction.

6. Discussion

The results explicitly refute the preregistered hypothesis in both Primary and Replication cohorts. The intervention (EntropyMaxWeightNoise) yielded lower held-out accuracy than deterministic baseline training on subject-disjoint UCI HAR data. This contradicts the expectation that entropy maximization of weight uncertainty parameters improves robustness in this setting.

These findings relate directly to prior work on Bayesian neural networks and robustness. *Tackling covariate shift with node-based Bayesian neural networks* states BNNs promise improved generalization under covariate shift [6]. However, our results suggest that explicitly maximizing entropy of multiplicative weight noise does not replicate this benefit in a PreNormTransformerEncoder on this dataset. This aligns with *Dangers of Bayesian Model Averaging under Covariate Shift*, which warns that approximate Bayesian inference considered robust can achieve poor performance under covariate shift [12]. While *A Diffractive Neural Network with Weight-Noise-Injection Training* proposes weight noise injection training to achieve strong robustness [14], our results indicate that entropy-maximized noise specifically may not generalize to subject-disjoint time-series classification in Transformers.

The negative results extend the understanding of uncertainty mechanisms in attention-based architectures. While *Alignment Entropy Regularization* uses entropy to measure a model's uncertainty in distributing probability mass [13], applying this to weight noise parameters did not improve out-of-distribution accuracy here. This suggests that uncertainty-inducing mechanisms optimized for diversity rather than posterior fidelity may not suffice for robustness in this domain.

****Scope Limitations****

We explicitly acknowledge the scope limitations of this testbed. The experiment used the UCI Human Activity Recognition dataset. While this provides a real-world subject-disjoint split, it is a specific domain of inertial time-series data. We prevent overgeneralization of these negative results to broader uncertainty quantification contexts; the failure of entropy maximization here does not necessarily imply failure in other modalities (e.g., vision, NLP) or under different shift types (e.g., corruption vs. subject-disjoint). The results validate or refute the core mechanism of Node-based BNNs [6] specifically within this architectural and data context.

7. Limitations

This study is limited to the UCI Human Activity Recognition dataset and a specific PreNormTransformerEncoder architecture. The negative results may not generalize to other data modalities or model sizes. The entropy maximization objective was applied only to the query projection layer; broader application might yield different results. Additionally, the study focuses on subject-disjoint shift, which is one form of distribution shift among many.

Declared scope. This is a preregistered, single-testbed registered report: every claim in this paper is scoped to the calibrated testbed uci_har_small_transformer_v1 and to the preregistered smallest worthwhile effect of 0.012509797746985206 proportion on held-out classification accuracy, a margin derived from the calibration's measurement noise (it reflects measurement precision, not

scientific importance). Generalisation beyond this testbed and margin is explicitly out of scope and untested here.

8. Conclusion

We conducted a Registered Report to test whether explicit entropy maximization of weight uncertainty parameters improves out-of-distribution accuracy on subject-disjoint time-series data. Using the UCI Human Activity Recognition dataset, we found that the intervention significantly worsened accuracy compared to deterministic baseline training in both Primary and Replication cohorts. The preregistered hypothesis was refuted. These results suggest that practitioners should not prioritize entropy maximization of weight noise for robustness tasks in this specific context over standard regularization. Future work should investigate whether posterior fidelity objectives or different uncertainty mechanisms perform better under subject-disjoint shift in Transformers.

Reproducibility

All results were produced by an automated pipeline running the experiment on GWDG HPC inside a container. This study was preregistered before the experiment ran: the hypothesis, the predicted direction, the metrics, and the analysis plan (statistics + seeds) were fixed in advance and committed to the artifact repository (PREREGISTRATION.md). Random seeds:

[497360171578835126, 619747419151856849, 4359311500859520721, 3900812323465083926, 1516445550462428226, 1630043136886007277, 1835401958502649079, 4224212089476219622, 2975544019134720152, 3507764674343915068, 4004663391355442268, 1509823930515868492, 3483967100743641674, 2225936250788509260, 1957754086644492888, 4154286735266072875, 3813494016017719200, 690472406820790188, 3813916678459859089, 4090465763900842853, 1844212296145829516, 3081923774287884327, 3477292413922411937, 726925812904865219, 4400454224498690158, 1309720931756809616, 3670000523134785215, 1572470746349086798, 5569990145947559, 2855297384752490100, 3166602034212987559, 4221459448664290905, 2934143650096726109, 1462463051419223747, 4363375170763905339, 3648570110071838823, 258115887333994485, 2427086170940110629, 4084426303853737760, 2996901096067535381, 2815367292871095378, 4593572677630054665, 3754669615394449950]. Slurm job id: 15540577. Container image: pytorch/pytorch@sha256:c8268a92a69bd500f8be0e665b2630ee006dadaf7bfbcb24249141b15ff622755. Walltime: 01:30:00. The exact code, container reference, seeds, run logs, Slurm job id, and the agent's full reasoning trace (including failed attempts) are in the artifact repository: (pending publication).

Ethics

This is a small-scale computational study using synthetic or small public data and no human subjects or personal data; it poses no foreseeable ethical or dual-use risks. Compute use was deliberately minimal.

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